Rational Functions & Asymptotes

Section 1: Vertical Asymptotes vs. Removable Discontinuities

1 Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 9}{x^2 - x - 6}$.

a. Vertical asymptote at $x = -2$; hole at $x = 3$

b. Vertical asymptote at $x = 3$; hole at $x = -2$

c. Vertical asymptotes at $x = 3$ and $x = -2$; no holes

d. Holes at $x = 3$ and $x = -2$; no vertical asymptotes

2 What is the coordinate (x, y) of the hole in the graph of $g(x) = \frac{2x^2 + 6x}{x^2 - 9}$?

a. $(-3, 1)$

b. $(-3, 0)$

c. $(3, 1)$

d. $(-3, \infty)$

3 If $h(x) = \frac{x - 4}{(x - 4)^2(x + 1)}$, which describes the behavior of h near $x = 4$?

a. $h(x)$ has a vertical asymptote at $x = 4$

b. $h(x)$ has a removable hole at $x = 4$

c. $h(x)$ has a horizontal tangent at $x = 4$

d. $h(x)$ is continuous at $x = 4$

Section 2: Horizontal Asymptotes and Degrees

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4 Determine the horizontal asymptote of $R(x) = \frac{5x^3 - 2x + 1}{2x^3 + 7x^2 - 4}$.

a. $y = \frac{5}{2}$

b. $y = 0$

c. $y = -\frac{1}{4}$

d. No horizontal asymptote

5 Which rational function has a horizontal asymptote at $y = 0$?

a. $f(x) = \frac{4x + 1}{3x^2 - 5}$

b. $g(x) = \frac{3x^2 - 5}{4x + 1}$

c. $h(x) = \frac{4x^2 + 1}{3x^2 - 5}$

d. $k(x) = \frac{4x^3}{3x^2 - 5}$

6 Can the graph of a rational function intersect its horizontal asymptote?

a. Yes, the graph can cross a horizontal asymptote at finite values of x

b. No, an asymptote can never be touched or crossed by definition

c. Only if the horizontal asymptote is $y = 0$

d. Only if the function has no vertical asymptotes

Section 3: Slant (Oblique) Asymptotes

7 Find the slant asymptote of $f(x) = \frac{2x^2 + 5x - 1}{x + 2}$.

a. $y = 2x + 1$

b. $y = 2x + 9$

c. $y = 2x$

d. $y = x + 1$

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8 Under what condition does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. The degree of $P(x)$ is exactly 1 greater than the degree of $Q(x)$

b. The degree of $P(x)$ equals the degree of $Q(x)$

c. The degree of $Q(x)$ is exactly 1 greater than the degree of $P(x)$

d. The degree of $P(x)$ is at least 2 greater than the degree of $Q(x)$

9 Find the equation of the slant asymptote for $g(x) = \frac{x^3 - 3x^2 + 4}{x^2 + 1}$.

a. $y = x - 3$

b. $y = x + 3$

c. $y = x$

d. $y = -3x$

Section 4: Limit Notation and Asymptotic Behavior

10 For $f(x) = \frac{1}{x - 3}$, what are $\lim_{x \to 3^+} f(x)$ and $\lim_{x \to 3^-} f(x)$?

a. $\lim_{x \to 3^+} f(x) = \infty$ and $\lim_{x \to 3^-} f(x) = -\infty$

b. $\lim_{x \to 3^+} f(x) = -\infty$ and $\lim_{x \to 3^-} f(x) = \infty$

c. Both limits equal ∞

d. Both limits equal $-\infty$

11 If $\lim_{x \to -\infty} f(x) = 4$, which statement is justified?

a. The line $y = 4$ is a horizontal asymptote of f

b. The line $x = 4$ is a vertical asymptote of f

c. $f(4)$ is undefined

d. The range of f is $[4, \infty)$

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12 Given $k(x) = \frac{2x-6}{(x-3)(x-5)}$, evaluate $\lim_{x \to 3} k(x)$.

a. -1

b. 1

c. ∞

d. 0

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